

Allamaprabhu Ani

London, United Kingdom, +44 774 174 6172, allamaprabhuani@gmail.com
allamaprabhuani.github.io | linkedin.com/in/allamaprabhu-ani | github.com/allamaprabhuani
scholar.google.com/citations?user=dX3xKGsAAAAJ

SUMMARY

PhD student in computational mechanics and machine learning, building [PhAST](#), a matrix-free, differentiable PyTorch solver for phase-field fracture. Lead author of the 2026 arXiv solver preprint and the 2026 *Engineering Fracture Mechanics* review on ML for fracture (with EPFL and University of Florida). Built validated aerospace hand-calculation automation tools in Excel-VBA.

SKILLS

Programming: Python, PyTorch (autograd, CUDA), JAX, NumPy/SciPy, scikit-learn, pandas, MATLAB, C++ (read), Bash, Git.

Machine Learning: Neural operators (DeepONet, FNO), PINNs, differentiable simulation, inverse problems, L-BFGS, adjoint methods.

Scientific Computing: GPU acceleration, sparse linear algebra, finite element method, phase field fracture, mixed precision numerics.

Infrastructure and tools: Slurm HPC, Linux, packaging, Sphinx, GitHub Actions, Akantu, FEniCSx, COMSOL, MSC Nastran/Patran, Abaqus.

EXPERIENCE

PhD Student, Computational Mechanics and ML Jul 2023 – Jun 2026 (expected)
City, St George's, University of London & Queen Mary University of London

- Built [PhAST](#), a matrix-free, differentiable PyTorch solver for phase-field fracture with end-to-end autograd through a staggered solver, scaling to 1M+ DOF on a single GPU.
- Implemented gradient-based inverse analysis for fracture-toughness recovery, spatial field reconstruction, and defect localisation; benchmarked against finite-difference and MCMC baselines.
- Achieved 10-20x sparse solve speedup over a C++ reference (Akantu) on quasi static benchmarks; cross validated dynamics on Kalthoff, SENT, Borden.
- Maintained reproducible Slurm pipelines on Hyperion2; local Mac CPU and WSL CUDA development.

Visiting Researcher, Computational Solid Mechanics 2025
LSMS Lab, EPFL (host: Prof. J.-F. Molinari)

- Lead author of the 2026 *Engineering Fracture Mechanics* review of ML for computational fracture; open access, growing citation count.

Founder and Tool Developer 2020 – 2021
Aeroknacks India Ltd

- Shipped aerospace hand-calculation automation tools for Excel-VBA and structural-analysis workflows, saving an estimated 10+ hours of manual calculation per week per engineer. Built from E.F. Bruhn, Michael Niu, Boeing Design Manuals, and ESDU references; included Cozzone, lug strength, fastener load transfer, buckling, modal, and ABD tools.

EDUCATION

PhD, Mechanical Engineering, City, St George's, University of London (Jul 2023 – Jun 2026 exp.). Thesis: matrix-free differentiable fracture mechanics and inverse analysis in PyTorch.

MTech, Design and Manufacturing, National Institute of Technology Silchar (2021–2023). CGPA 9.88/10. Best Student Award.

BE, Mechanical Engineering, GM Institute of Technology, VTU (2016–2020). CGPA 8.47/10.

SELECTED PUBLICATIONS

Ani, A.S., Molinari, J.-F., Subhash, G., Ponnusami, S.A. (2026). A matrix-free, differentiable PyTorch solver for phase-field fracture: Formulation, benchmarks, and inverse analysis. [arXiv:2606.23458](#).

Ani, A.S., Nakka, R., Subhash, G., Molinari, J.-F., Ponnusami, S.A. (2026). Machine learning for computational fracture and damage mechanics: Status and perspectives. *Engineering Fracture Mechanics* 332, 111778.

PROJECTS

PhAST – github.com/CEMS-Lab/PhAST; matrix-free differentiable PyTorch software for phase-field fracture.

AWARDS

Yeoman & Travelling Scholarship (2026, 1 of 3 citywide); Dean's Award for Outstanding Teaching Support (2026); AFHEA (2024); Funded PhD Studentship (2023); Best Student Award (2023).